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COMMENTARY

THE HOW AND WHY OF THE MOVEMENT SYSTEM AS THE IDENTITY OF PHYSICAL THERAPY

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ABSTRACT

The Movement System was adopted as the identity of physical therapy as one of the 8 guiding principles accompanying the Vision Statement of 2013. At its inception physical therapy was considered more of a technical field rather than that of a professional field. Physicians were to diagnose the patient's problem and the therapist was to follow the prescription provided by the physician with the primary purpose being to relieve symptoms such as pain or muscle weakness. Even by the 1960's, the prescription became more of a referral and there was recognition that therapists were making decisions about the patient's treatment and discharge disposition. The role of the physical therapist in pathokinesiologic problems has been well accepted over the years but as insights are gained about the role of movement in musculoskeletal pain, the concept of kinesiopathologic problems is being defined. Whether the movement dysfunction is from a pathokinesiologic or a kinesiopathologic mechanisms, the underlying physiologic process is movement which is the composite action of the movement system. This article provides a brief discussion of the steps leading to promotion of the identity and the reasons that further defining and promoting the movement system as the body system for which physical therapists are responsible is necessary for the full recognition of the profession. As suggested by the kinesiopathologic concept of movement inducing pathology, physical therapists can address the cause of musculoskeletal problems and not just symptoms or consequences such as the pathoanatomic problem.

Level of Evidence: 5

Key words: Identity, movement system, pathokinesiologic, kinesiopathologic

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THE STEPS LEADING TO AN IDENTITY

More than four years have passed since the American Physical Therapy Association (APTA) designated the Movement System (MS) as the identity of physical therapy.[1] In the process of developing a vision statement that addressed what physical therapy can do for society, the insightful members of the committee enumerated eight principles.[2] The first among these being an identity. The resolution on identity states:

> The physical therapy profession will define and promote the movement system as the foundation for optimizing movement. The recognition and validation of the movement system is essential to fully understand the physiological function and potential of the human body. The profession will be responsible for monitoring an individual's movement system across the life span in order to promote optimal development, diagnose dysfunction, and provide interventions targeted at preventing or ameliorating restrictions to activity and participation. The movement system will form the basis of practice, education, and research of the profession.

Amazingly but probably not surprisingly, the issue of an identity had not been specifically addressed in the 96 years of existence of the APTA, though over the years some leaders of the profession have raised the issue.[3,4,5] Why the delay? Did the profession already have an identity? If so, what is the identity? From where did it emanate? Is the commonly assumed identity accurate? Was there agreement on the previously assumed identity? A year after the House of Delegates approved the vision statement and the eight guiding principles, an article was published providing an overview of the historical development of the MS and steps by the APTA and by individuals to develop the concept.[6] This article discussed a more general perspective of changes in physical therapy and medicine over the years that lead to the need to develop a clear identity that reflects the extensive changes in the profession over the past 90 years.

IMPLICATIONS OF VISION 2020 AFFECTING IDENTITY

The perspective of the current changes is derived from 60 years in the profession and the opportunity to observe and participate in an amazing transformation of physical therapy. However, this transformation has occurred primarily within the profession and is not widely evident to those on the outside, such as other health professions and the public. The early (but never specifically stated) identity of physical therapy was that of a technical field with clinicians providing physical treatments prescribed by physicians. What a contrast to the goal of the past two vision statements! In Vision 2020 adopted in 2000, the profession's stated lofty goals were consistent with recognition of an autonomous health provider.[7] Vision 2020 stated the goal of the profession to be doctors, diagnosticians, and have direct access to provision of care for patients. That is a marked change from what started as a technical field. Physical therapy is certainly recognized as a profession but at what level? How far has our currently assumed identity progressed? The questions raised by vision 2020 include:

- How does having a doctoral degree differ from having a master's degree?
- What are we going to diagnose?
- What do we offer the public as direct access practitioners?
- What is expected from doctors?

How does having a doctoral degree differ from having a master's degree? The primary rationale at the time of raising entry level to the clinical doctorate was that if patients have direct access to physical therapy, we must be able to recognize conditions that do not "belong" to us. Though certainly a necessary and an admirable role that was always our responsibility. But that rationale does not address what we do for the conditions that ***do*** belong to us.

What are we going to diagnose? Just making a statement that we are diagnosticians without providing specific conditions with specific labels makes the claim rather hollow. The lack of diagnostic labels detracts from recognition that we can determine cause or defining characteristics of the conditions we treat. Furthermore, the lack of diagnostic labels fails to convey that movement impairments are not just problems of muscle weakness but can and need to be defined as syndromes. If neurologists or cardiologists claimed to be diagnosticians but did not have specific labels to identify well described pathological

conditions, they would not have the respect of the public or other health practitioners that exists today. The lack of well described diagnostic conditions would also impact treatment and pursuit of underlying pathophysiology. Physical therapy has not followed the pattern of physicians, who classified and described pathophysiological and pathoanatomical conditions, by doing the same with pathokinesiologic and kinesiopathologic conditions.

Why should we have direct access? What do we offer that sends a clear message to the public that they should consult us directly? A common current public belief is that if a health condition, such as a pain problem, is present then a diagnosis is necessary. Are we in any way conveying to the public that a physical therapist can "diagnose" your problem, that there is a label for the condition and a specific method for treating the condition? The prevailing belief is that the doctor diagnoses the problem and if the doctor believes the physical therapist can help relieve the "symptoms" not the problem, the patient will be sent to a therapist. The failure to clearly define what a physical therapist offers, which is directly related to an identity is a major impediment to reaping the benefits of what was implied by vision 2020. Our new vision of how we are going to use our "expertise" to transform society by optimizing movement to enhance the human experience, is going to remain a wishful statement if we do not make the difficult decisions and take the necessary steps to implement all that is implied by designating the movement system as our identity, our expertise, and our responsibility.

TRANSITION FROM TECHNICIAN TO PROFESSIONAL

The status of physical therapy 60 years ago provides some insight into why the transformation of our profession has been so difficult. Entering the field at the end of the polio era, my expectations were that I would help paralyzed children regain their ability to move their limbs and be mobile. My two years of basic college courses were intended to provide enough of a science background to understand anatomy, kinesiology, and physiology. With that information, when the doctor told me what to do, I would understand enough not to be dangerous and maybe even to have some insight into what was happening when stretching muscles to prevent contractures, using muscle re-education to restore muscle function, and practicing basic mobility activities. Of course, manual muscle testing was an important component so that the physician could use the results for identifying what neuronal pathways had been affected by the disease. Manual muscle testing also gave us the information to set expectations for what functional outcome could be expected and what kind of braces and assistive devices would be necessary. During this period, the physical therapist had a clear role with the paralyzed patient of therapeutic exercise to restore muscle function and functional performance. The "prescription" from the physician was very nonspecific, just a diagnosis and therapeutic exercise. In fact, Worthingham[8] indicated that by the 1970's often the diagnosis was not included in the "prescription" from the physician.

The patient with musculoskeletal pain was a different situation. The "prescription" was usually very specific, detailing the modalities to be used and even some specificity for the exercises. At the time, stretching and strengthening along with modalities were the primary treatments because range of motion and adequate muscle strength seemed to be all that was necessary to restore movement and alleviate pain. Just think how many document systems and insurance companies still require this same information though we know well that this information has little to do with movement performance. We were not expected to figure out the underlying problem, but just implement the prescribed treatments. In fact, Medicare did not pay for physical therapy evaluation until the 1990's because the physician had performed the examination.

My education really did not address the type of decisions that were necessary for the treatment of the patient with upper motor neuron lesions. The limitation of the physician's diagnosis became particularly evident when patients with head injury, spinal cord injury, and stroke became the prevailing population in the clinic. What a challenge it was as we tried to understand the difference between treating the patient with upper motor neuron lesions versus those with lower motor neuron lesions. The physician's diagnosis (or even knowledge) was not much help. In fact, I was part of a study by the Department

of Labor that observed us working with patients and determined we made decisions and therefore could be labeled as professionals rather than technicians. I just wished I had a better understanding of the disordered motor control so the decisions were well founded. The recognition of the expertise of the physical therapist in the treatment of the paralyzed patient is well accepted. But these patients have a diagnosis made by the physician though that diagnosis does not provide useful information regarding the pathokinesiologic condition. The profession did not reinforce the decision- making capabilities of the physical therapist when most forms for referral used by physical therapy services listed the treatment methods and modalities and the physician was to check off what was to be done. Hopefully such forms are no longer used by physical therapy services.

TRANSITION IN PHYSICIAN EDUCATION

In physical therapy, particularly during the 1970's and 80's, many things were changing in education and practice as we slowly and subtly began to assume increasing responsibility for making decisions about patient management. Change was also occurring in the education and practice of the physician. In the 1950's and the early 60's medical education and practice were anatomically based. The medical student spent extensive hours in the anatomy lab performing dissection of the entire body. Many diagnoses were made based on the physical exam and the location of the pain. Then in the late 60's, the hours in anatomy were reduced and the increased time was spent in biochemistry courses. Currently in some medical schools only about one week is spent on dissection of the limbs and in others about two weeks. Kinesiology is not part of the course work. Now the educational emphasis for the medical student is cell biology and genetics. In many medical schools, the departments of anatomy have been replaced by neurobiology and physiology by cell biology and physiology. At Washington University School of Medicine St. Louis, anatomy is taught by anthropologist because of the lack of PhDs in anatomy which is reflective of the lack of research related to anatomy. Even systems physiology is not a common focus of research in most basic science departments. As is well known, the physical exam is a small component of the physician's exam with the primary emphasis being on laboratory and radiological tests. The other major change has been the extensive development of specialists. Though there are many advantages to having specialists in a specific body region, such as the hip reconstruction surgeon, the disadvantage is limited recognition of the interaction of multiple body systems and regional interactions. Thus, a major change over the years is that the physician has minimal knowledge of muscle and movement function. Anatomy and the physical exam are not a major part of practice. Not only is the physician knowledge limited in these areas but there is also limited appreciation for the importance of movement and the role of contributing systems. Recognition of what has changed in physical therapy education and practice is important but so too is recognition of what has changed in the education and practice of the physician.

RECOGNIZING THE ROLE OF MOVEMENT AND THE NEED FOR A PROFESSION-GENERATED IDENTITY

As indicated in the prior information, physical therapists were initially considered to be technicians providing treatment prescribed by the physician. Changing the perception of a health field is more difficult than establishing the desired perception at inception. The earliest definition of physical therapy was treatment by physical means, that included physical agents or modalities and exercise. Of course, the value of this treatment was to restore movement of the limbs and general mobility. So, physical therapy was about movement but on a very simplistic basis of having adequate muscle strength and joint range of motion. The movement disorders of patients with upper motor neuron lesions clearly demonstrated that movement was the result of multiple complex processes. With musculoskeletal pain problems, the development of manipulative and manual therapy reflected the limited effectiveness of strengthening and stretching. The emphasis was then on correcting joint motions rather than limb movements. But the use of movement, even passively applied, was the mode of treatment. What this era did initiate, was the expectation that the therapist should determine the cause of the pain, a relatively new concept.

At the same time, physical therapists were obtaining PhD's and performing scientific studies. When

physical therapy education was at the baccalaureate level with only two years of basic education, entering a PhD program was not possible without additional education. As more of the physical therapy students had a bachelor's degree before their professional studies, a larger cadre of individuals were qualified and interested in pursuing research degrees. These individuals recognized the need for defining our body of knowledge or underlying science. Helen Hislop identified this need in her 1975 Mary McMillan Lecture and suggested that pathokinesiology is the underlying science.[3] She also stated that the profession suffered from a lack of an identity. Steve Rose, PhD, PT, FAPTA was also an advocate of promoting pathokinesiology as the defining science and even suggested changing the name of the profession.[9] He was also a major promoter of patient classification based on a physical examination that serve as a diagnosis.[10] During a 1984 conference designed to discuss the concept of pathokinesiology, Jules Rothstein stated that 10 years after Hislop reported the problem of the profession's identity the problem is no better.[5] As Dr. Rothstein stated, **we are known for what we do and not for what we know.** That situation persists today. Several papers were published in the early 1990's as the outcome of a meeting of academics and clinicians who were interested in furthering the prescient ideas of Dr. Rose. One of the recommendations was to enlarge on the role of physical therapy from just pathokinesiology to include optimization and prevention related to **movement**. Promoting the concept of movement science was believed to be the optimal way of addressing the comprehensive scope of the body of knowledge of physical therapy. But having a science does not specifically address the clinical application of that science. Just as having neuroscientists does not address the role of neurologists. Describing a movement system, a physiological system of the body, as the focus for clinical practice would be consistent with the pattern used by physicians so successfully.

The evolution in physical therapy education and practice has in many ways paralleled the evolution of understanding movement. Previously adequate interventions targeted at return of muscle strength and range of motion were considered all that were necessary to restore movement and functional performance. The disordered movement resulting from central nervous system lesions demonstrated that the production and regulation of movement is highly complicated. Few physicians were interested in the specifics of the movement disorder and were more interested in the nervous system lesion itself and what type of medication could be used to alleviate symptoms. More recently the role of motor control in patients with musculoskeletal disorders has been the focus of many studies. Only a few orthopaedic surgeons have begun to recognize the role of movement as an important factor in musculoskeletal pain. Beginning in the 1980's physical therapists were becoming scientists and studying movement disorders as well as aspects of musculoskeletal pain and exercise physiology. An indication of the quality and recognition of research by physical therapists was in the grants provided by the National Center for Medical Rehabilitation and Research (NCMRR). Though designed to promote research by physiatrists, within a few years of its founding, the majority of grants were given to physical therapy based PhD programs. As science was growing, clinical practice remained highly variable in approaches to patient care and was also subjected to demands for productivity. Physicians had minimal background in anatomy and none in kinesiology, so their focus became the underlying pathoanatomical problem now revealed by advances in radiology. Symptom relief became the major emphasis which escalated from anti-inflammatory treatment to forms of opiates for pain relief. The prevailing belief being that if the symptoms subside the condition is resolved, thus movement and activity can be resumed. The inevitability of musculoskeletal pain and osteoarthritis was accepted. What failed to be asked, was how is movement contributing to the problem and how could movement be optimized to slow the process just as activity slows the onset of diabetes and cardiovascular disorders?

Clearly physical therapists were gaining in knowledge about movement-related disorders and ways in which movement could be used for treatment. These gains were reflected in the expansion of the educational content and in the philosophic position adopted by the 1983 House of Delegates[12]

> Physical therapy is a health profession whose primary purpose is the promotion of optimal human health and function through the application of scientific principles to assess, correct, or alleviate acute or prolonged movement dysfunction.

Subsequent modifications of this position in the Guide to Physical Therapist Practice and adopted by the House of Delegates in 1999, expanded the statement of processes used,

> "to include examination, evaluation, diagnosis, prognosis and intervention to prevent or remediate impairments in body structures and function, activity limitations, participation restrictions or environmental barriers as related to movement and health".[13]

CONSIDERATIONS OF THE PROFESSION'S BUSINESS MODEL AND EFFECT ON THE PRACTICE MODEL

Thus, statements have been made about physical therapy being a doctoring profession but the critical components underlying that status have not been implemented. In part, this can be attributed to the relative business success of physical therapy as indicated by the expansion of educational programs, both for the physical therapist and for the physical therapist assistant, and by profitable practices. When practice is perceived to be good because of profitability, there is little motivation to change or to assess it's weaknesses. The enactment of Diagnostic Related Groups (DRGS) had a substantial impact on physical therapy in the hospital. No longer was physical therapy a source of substantial revenue. Services moved to outpatient clinics that were not constrained by a fixed amount of money to cover services. So, the effect of DRGs was only temporary but during that time therapists did lose jobs and other sources of revenue had to be found, such as outpatient services and home health. Then the balanced budget act of 1997, which implemented the cap on physical therapy services covered by Medicare had a serious impact on both education and practice. Therapists lost their jobs, salaries went down 20%, applications and enrollment in physical therapy programs were dramatically reduced. Modification of the cap and restructured provision of services restored financial viability mostly by increasing productivity. But there are many reasons to anticipate a not so distant challenge to the business of physical therapy as cost containment becomes a major emphasis by the government and insurance carriers. As the profitability of our previous business model is challenged, there is a necessity to make a strong and vivid case for what physical therapy offers that is not provided by any other profession. Practice based on optimizing movement and the performance of the systems constituting the MS would 1) contribute to maintaining health, and 2) to cost containment by reducing drug and surgical interventions. Recognition of the MS as an important physiological system by the public and other health professions, the use of MS diagnoses, promotion of yearly examinations, lifespan practice, and an emphasis on prevention of movement impairments would provide the specifics for achieving the objectives of vision 2013. Our practice model must change, which requires a clear statement of identity and implementation of all that is implicit in that identity.

WHY THE MOVEMENT SYSTEM AND NOT JUST MOVEMENT?

The group that gathered to further the ideas expressed by Dr. Rose not only suggested the promotion of movement science but were also the first to suggest the idea of the movement system (MS). Over the ensuing 27 years, the concepts have been clarified and developed to varying extents by the approximately 20 individuals in attendance. The Diagnosis Dialog group also reinforced the idea of the movement system. Why use the term MS and not just movement? One reason is that in health care, practice is centered around a body system, which speaks to the expertise of the practitioner. The neurologist is recognized as the expert in the nervous system and the cardiologist as the expert in the cardiovascular system. Though these examples address key anatomical systems and their physiology or their pathophysiology, other physiological processes that involve multiple systems have also been identified as areas of specific practice. For example, the endocrinologist is responsible for the metabolic system, and the immunologist for the immune system. These are systems of systems just as the MS is a system of organ systems as depicted in Figure 1. Movement is a key physiological process

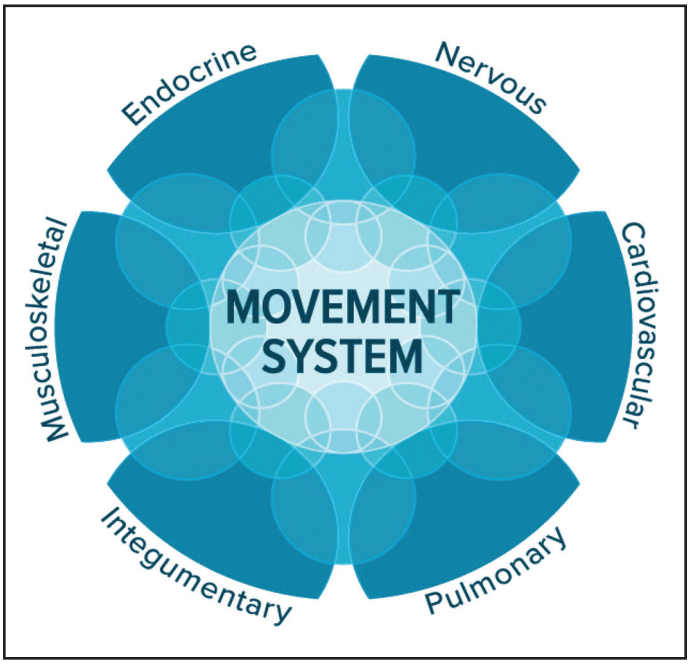

Figure 1. *The movement system adopted by the APTA Board of Directors 2016. Used by permission of the APTA.*

that is essential at all levels of the organism from subcellular to joint, limb, and total body movement. If movement was the only focus of physical therapy, that would limit our examinations and treatment to just movement. That would not convey our in-depth knowledge of the musculoskeletal, nervous, cardiovascular, endocrine, and pulmonary systems that is necessary to understand movement and movement impairments.

Another important aspect of the MS is that guidance for growth and optimal development can become a major responsibility of the physical therapist. Almost daily, evidence is increasing that lifestyle is the major contributor to disease and disability. For example, inactivity has been cited as contributing to a variety of cancers, as well as to the metabolic syndrome a forerunner to diabetes and cardiovascular disease. A recent publication has suggested the major increase in knee joint osteoarthritis is from the reduction in physical activity associated with the post-industrial age.[14] What an irony that as a society we consult our dentist from once to twice a year from a very early age and for the rest of our lives when the major contributions of our oral cavity is for speaking and eating. Yet our body, our MS, is needed for everything and an expert is not consulted for guidance in development and for prevention. Thousands of dollars are spent to straighten our teeth which is primarily for appearance but no attention is paid to alignment and movement patterns of the body which are essential to all activity.

Movement is key and optimizing the contributions of all the participating organ systems and movement itself can reasonably defer the onset of osteoarthritis as well as other system diseases. In many ways physicians are forced to treat symptoms and consequences while physical therapists can diagnose and treat cause and better yet, play a role in prevention. Conservative treatment, reducing the need for drugs and surgery is cost saving and optimizing the human experience. To be appreciated is that conveying to other health care practitioners and to the public that "therapists" can diagnosis and are responsible for a system of the body is a challenge just based on terminology. As the speech therapists have changed their name to speech pathologists, so should the physical therapy profession consider a change in name that would be in keeping with a change in practice responsibilities.

WHAT IS NEXT?

Upon designating the MS as the identity of physical therapy, the APTA appointed a Board Work Group to develop a definition and a plan to disseminate and implement the MS in practice, education and research. A comprehensive plan was developed as well as a white paper that was published by the APTA and a recommendation for a MS summit. A task force was then appointed to plan and conduct the summit. The task force was to further develop the plan for dissemination and implementation. The summit that was held in December 2016 was considered a success by all those who participated. Three main topics were discussed. The first was terminology, the second was developing a basic movement system exam with a focus on movement, and the third was on diagnosis. The recommendations from the summit were submitted to the APTA Board of Directors. The Board approval for implementation of the recommendations is based on numerous factors including the budget and mandates from the House of Delegates. Though specific steps in enacting the recommendations will be forthcoming, there is no need for physical therapists to wait on promoting the MS. Using the term

with patients and in promotional materials is going to be necessary. Implementing practices that provide yearly examinations based on an assessment of the MS would help convey the idea of the importance of optimal movement and that the physical therapist has the expertise that is required. Nothing is more important than movement when considered at all levels of the organism, from ions going through membranes, the flow of synaptic transmitter, how the limbs move, how body segments interact, to how we move in our environment. Just as eating, such an intrinsic part of life, has been taken for granted until recent years, so has movement been taken for granted. As we all know from multiple sources, appropriate nutrition is considered a major factor in health. So must we make the public and other health care practitioners realize that optimal movement is probably even more complicated than optimal nutrition, and that we are the professionals that can provide the guidance that is necessary to be sure exercise and activity contribute to health and not to injury.

CONCLUSION

The APTA has recognized the need for an identity that is commensurate with the extensive changes in the profession that have occurred over the past 30 years. There is little recognition by the public or even by other health professionals that the educational requirement is seven years and that graduates are clinical doctors with extensive knowledge of all the systems that comprise the MS. Promoting the importance of a MS is not only the right thing to do but is also necessary. Knowledge of the movement system can highlight how movement induces pathology, that MS diagnoses can provide labels representing this process, and that a yearly MS exam can contribute to prevention of musculoskeletal problems. Because all indications are that our business model is going to change, physical therapists need to be pro-active and direct change in our practice model.

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